$A = (\{q_0, q_1, q_2\}, \{a, b, c\}, \{a, Z_0\}, \{a, Z_0\}, \delta, q_0, Z_0, \phi)$.

for 1st stack ↑ 2nd stack ↑ topmost symb. for both stacks. ↑

18/April/18

UNIT - 5 TURING MACHINE

i) The turing machine is a simple mathematical model of general purpose computer.

ii) Turing machine is capable of performing any calculation which can be performed by any computing machine.

iii) The most serious problem with finite state machine is that it has limited memory and hence it is inadequate model for the computer.

Turing machine resolves such problems and it is another class of models for computing machines.

iv) Turning machine was named after the scientist Alan M. Turing.

Turing Machine Model

1) The turing machine can be thought of as finite control connected to a R/W (Read / Write) heads;

ii) It has one tape which is divided into a number of cells.

iii) The block diagram of the basic model for the turing machine is given as follows:

Scanned with CamScanner

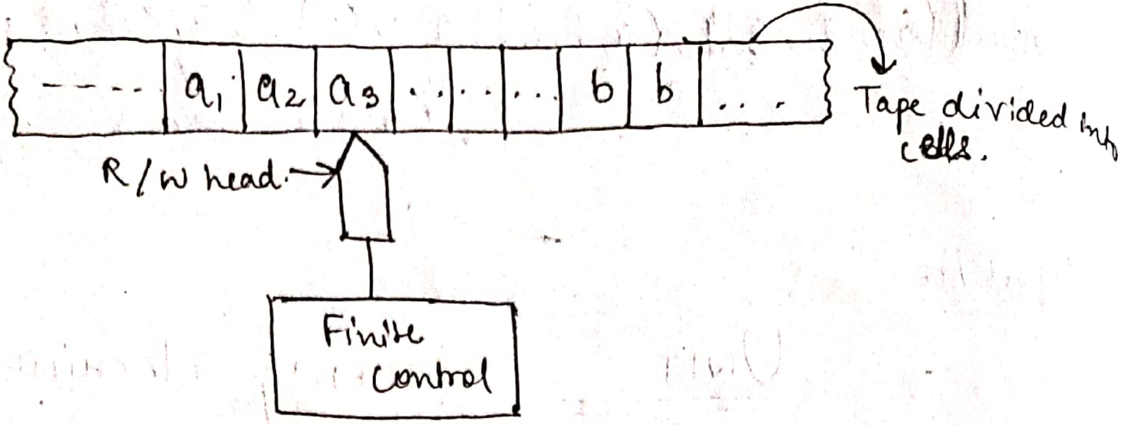


fig: Turing machine model

iv) Each cell can store only one symbol.

v) The input to and the output from the finite state automaton are effected by the R/W head which can examine one cell at a time.

vi) In one move, the machine examines the present symbol under the R/W head on the tape and the present state of an automaton to determine

a) A new symbol to be written on the tape in the cell under the R/W head.

b) A motion of the R/W head along the tape: either the head moves one cell left (L), or one cell right (R)

c) The next state of the automaton.

d) Whether to halt or not.

Defn of Turing Machine :-

A turing machine is defined by 7 tuple given as $M = (Q, \Sigma, \Gamma, \delta, q_0, b, F)$ where.

Q is a finite non-empty set of states.

Γ is a finite non-empty set of tape symbols.

Scanned with CamScanner

$b \in \Gamma$ is a blank.

Σ is a non-empty set of input symbols and $b \notin \Sigma$

$q_0 \in Q$ is the initial state.

$F \subseteq Q$ is the set of final states

δ is the transition function such that δ is.

$$Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$$

Current state of the machine and current tape symbol being read. The result is a new state of the machine, a new tape symbol, which replaces old one and a move symbol L or R i.e., $\delta(q, x) = \{(q', y, L)\}$

* NOTE :-

i) The acceptability of a string is decided by the reachability from the initial state to some final state. So the final states are also called the accepting states.

ii) δ may not be defined for some elements of $Q \times \Gamma$

Representation of Turing machine

We can represent turing machine by :

i) Instantaneous description (ID) using move relation.

ii) Transition Table.

iii) Transition Diagram

19/April

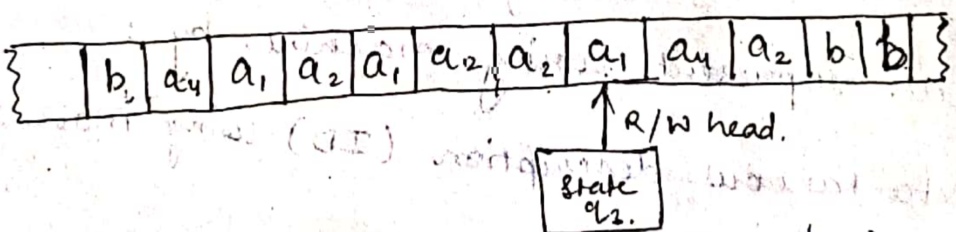
→ Representation by Instantaneous Description (ID)

i) 'Snapshots' of a turing machine in action can be used to describe a turing machine. These give 'instantaneous descriptions' of a turing machine.

ii) An ID of a turing machine is defined in terms of the entire input string and the current state.

Defn of ID: An ID of a turing machine M is a string ~~αβα~~ $\alpha\beta\gamma$, where β is the present state of M, the entire input string is split as $\alpha\gamma$, the first symbol of γ is the current symbol 'a' under the R/W head and γ has all the subsequent symbols of the input string, and the string α is the substring of the input string formed by all the symbols to the left of 'a'.

Example: A snapshot of turing machine is shown in the figure below. Obtain the ID.



Soln: The present symbol under the R/W head is 'a_1'.

The present state is q_3.

So, a_1 is written to the right of q_3.

The non-blank symbol to the left of a_1 formed the string $a_4 a_1 a_2 a_1 a_2 a_2$, which is written to the left of q_3.

The sequence of non-blank symbols to the right of a_1 is $a_4 a_2$

Thus, the ID is given as.

$\alpha \beta \gamma$

$\boxed{a_4 a_1 a_2 a_1 a_2 a_2 q_3 a_1 a_4 a_2.}$

*NOTE :-

i) For constructing the ID, we simply insert the current state in the input string to the left of the symbol under the R/W head.

ii) We observe that the blank symbol may occur as part of the left or right substring.

Moves in a TM :

* Suppose $\delta(q, x_i) = (p, y, L)$
The input string to be processed is $x_1 x_2 \ldots x_n$ and the present symbol under the R/W head is x_i.
So, the ~~ID~~ move in ID is given as $x_1, x_2 \ldots x_{i-2} x_{i-1} q x_i x_{i+1} x_n$

$\vdash x_1, x_2 \ldots x_{i-2}\, p\, x_{i-1}\, y\, x_{i+1} \ldots x_n$

* If $\delta(q, x_i) = (p, y, R)$
The input string to be processed is $x_1 x_2 \ldots x_n$ an the present symbol under R/W head is x_i
So, the move in ID is given as

$x_1 x_2 \ldots x_{i-1}\, q\, x_i\, x_{i+1} \ldots x_n.$

~~$\vdash x_1, x_2 \ldots x_{i-1}\, y\, p\, x_{i+1}\, y\, x_{i+2} \ldots x_n$~~

$\vdash x_1 x_2 \ldots x_{i-1}\, y\, p\, x_{i+1} \ldots x_n.$

→ Representation by Transition Table

i) We give the definition of δ in the form of a table called transition table.

ii) If $\delta(q,a) = (\gamma, \alpha, \beta)$, we write $\alpha\beta\gamma$ under the α-column and in the q-row

iii) So, if we get $\alpha\beta\gamma$ in the table, it means it α is written in the current cell, β gives the movement of the head (L or. R) and γ denotes the new state into which the turing machine enters.

iv) For example : Let us consider a turing machine with 5 states $q_1, q_2 \ldots q_5$, where q_1 is the initial state and q_5 is the only final state, the tape symbols are 0, 1, b. The transition table given below describes δ.

α-column

Present State	Tape Symbol		
	b	0	1.
$\rightarrow q_1$	$1Lq_2$	$0Rq_1$	—
q_2	bRq_3	$0Lq_2$	$1Lq_2$
q_3	~~$0Rq_5$~~ —	bRq_4	bRq_5
q_4	~~$0Lq_2$~~ $0Rq_5$	$0Rq_4$	$1Rq_4$
q_5	$0Lq_2$	—	—

q-rows ↓

Example : Consider the turing machine description given in the above table. Draw the computation sequence of the input string 00.

Soln: For the input string 00b., we get the following sequence :

$q_1 00b \vdash 0q_1 0b$

$\vdash 00q_1 b$

$\vdash 0q_2 01$

$\vdash b q_2 001$ $\qquad\qquad q_2 001$

$\vdash q_2 b001$ $\qquad\qquad \vdash$

$\vdash bq_3 001$

$\vdash bbq_4 01$

$\vdash bb0q_4 1$

$\vdash bb01q_4 b$

$\vdash bb010q_5 b$

$\vdash bb01q_2 00$

⋮

Since q_5 is the final state. So, the given string is accepted by the turing machine in the previous step.
The further steps may enhance the transition movements to get some more states before reaching again to the final state.

Scanned with CamScanner

Example :

Present State	Tape Symbol b	0	1
$\rightarrow q_0$	—	$1Rq_1$	$1Rq_0$
q_1	bLq_2	—	$1Rq_1$
q_2	—	—	$0Lq_3$
q_3	bRq_4	—	$1Lq_3$
q_4	~~bRq_4~~	—	—

Draw the computation sequence of the input string 111011

$q_0\,111011b \vdash 1q_0\,11011b$

$\vdash 11q_0\,1011b$

$\vdash 111q_0\,011b$

$\vdash 1111q_1\,11b$

$\vdash 11111q_1\,1b$

$\vdash 111111q_1\,b$

$\vdash 11111q_2\,1b$

$\vdash 1111q_3\,10b$

$\vdash 111q_3\,110b$

$\vdash 11q_3\,1110b$

$\vdash 1q_3\,11110b$

$\vdash bq_3\,111110b$

$\vdash q_3\,b111110b \vdash bq_4\,111110b$

Here there is no transition available for $q_4 1$, so the turing machine M Halts.
Since q_4 is the final state so the string is accepted by turing machine.

<u>Example</u> Consider the turing machine M described by the transition table given below. Describe the processing of
(a) 011 (b) 0011 (c) 001 using
ID's, which of the above strings are accepted by M.

Present State	Tape Symbol 0	1	x	y	b.
$\rightarrow q_1$	xRq_2	—	—	—	bRq_5
q_2	$0Rq_2$	yLq_3	—	yRq_2	—
q_3	$0Lq_4$	—	xRq_5	yLq_3	—
q_4	$0Lq_4$	—	xRq_1	~~$yxRq_5$~~ yRq_5	~~bRq_6~~ bRq_6
q_5	—	—	—	—	—
q_6	—	—	—	—	—

a) $q_1 011b \vdash x q_2 11b$
$\vdash b q_3 x y 1b.$
$\vdash b x q_5 y 1b.$
$\vdash b x y q_5 1b.$

M halts.
So, the input string is not accepted
∵ q_5 is not final state.

(b) $q_1 0011b \vdash x q_2 011b$
$\vdash x 0 q_2 11b$
$\vdash x q_3 0 y 1b$
$\vdash q_4 x 0 y 1b.$
$\vdash x q_1 0 y 1b.$
$\vdash x x q_2 y 1b$
$\vdash x x y q_2 1b$
$\vdash x x q_3 y y b.$
$\vdash x q_3 x y y b$
$\vdash x x q_5 y y b.$
$\vdash x x y q_5 y b.$
$\vdash x x y y q_5 b.$
$\vdash x x y y b q_6$

TM again halts. but string is accepted.

Scanned with CamScanner

c) $q_1\, 001b \vdash x q_2 01b$

$\vdash x0q_2 1b$

$\vdash x q_3 0yb$

$\vdash q_4 x0yb.$

$\vdash x q_1 0yb.$

$\vdash xx q_2 yb.$

$\vdash xxy q_2 b.$

$\not\vdash$

M halts as there is no transition and string is also not accepted as q_2 is not final state.

⇒ Representation by transition Diagram.

i) We can use the transition diagrams to represent turing machines.

ii) The states are represented by vertices.

iii) Directed edges are used to represent transition of states.

iv) The labels are triples of the form (α, β, γ), where $\alpha, \beta \in \Gamma$ and $\gamma \in \{L, R\}$

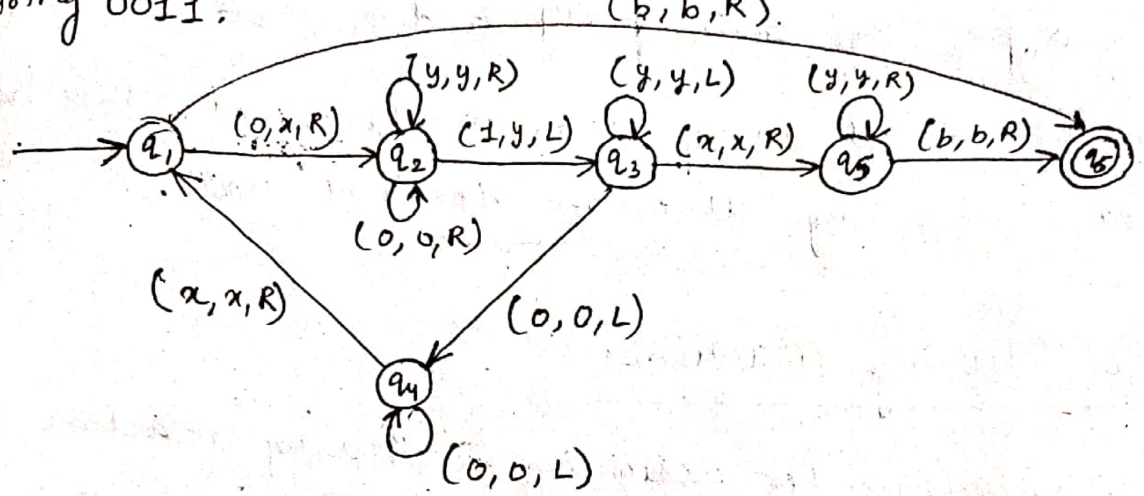
v) When there is a directed edge from q_i to q_j with label (α, β, γ), it means that $\delta(q_i, \alpha) = (q_j, \beta, \gamma)$

vi) Every edge in the transition system can be represented by a 5-tuple. $(q_i, \alpha, \beta, \gamma, q_j)$.

vii) So, each turing machine can be described by the sequence of 5-tuples representing all the directed edges.

viii) The initial state is indicated by → and any final state is marked with ◯.

Example: M is a turing machine represented by the transition system given in the figure below. Obtain the computation sequence of M for processing the input string 0011.



~~$q_1 0011b$~~ $q_1 0011b \vdash x q_2 011b$

$\vdash x0q_2 11b$

$\vdash x q_3 0 y 1 b$

$\vdash q_4 x 0 y 1 b$

$\vdash x q_1 0 y 1 b$

$\vdash x x q_2 y 1 b$

$\vdash x x y q_2 1 b$

$\vdash x x q_3 y y b$

$\vdash x q_3 x y y b$

$\vdash x x q_5 y y b$

$\vdash x x y q_5 y b$

$\vdash x x y y q_5 b$

$\vdash x x y y b q_6$

M halts as there is no transition from q_6 but ~~and~~ ~~also~~ string is ~~not~~ accepted as q_6 is final state.

Scanned with CamScanner

Language acceptability by Turing Machines

Let us consider the turing machine $M = (Q, \Sigma, \Gamma, \delta, q_0, b, F)$
A string w in Σ^* is said to be accepted by M if
$q_0 w \vdash^* \alpha_1 p \alpha_2$ for some $p \in F$ and $\alpha_1, \alpha_2 \in \Gamma^*$.

M does not accept w if the machine M either halts in a non-accepting state or does not halt.

Design of Turing Machines.

The basic guidelines for designing a turing machine is given as

1. The fundamental objective in scanning a symbol by the R/W head is to 'know' what to do in future. The machine must remember the past symbols scanned. The turing machine can remember this by going to the next unique state.
2. The number of states must be minimised. This can be achieved by changing the states only when there is a change in the written symbol or when there is a change in the movement of the R/W head.

Example: Design a turing machine to recognize all strings consisting of an even no. of 1's.

Sol^n: The construction is made by defining moves in the following manner:

a) q_1 is the initial state. M enters the state q_2 on scanning 1, and writes b

b) If M is in state q_2 and scans 1, it enters q_1 and writes b.

c) q_1 is the only accepting state. So M accepts the string if it exhaust all the input symbol and finally is in state q_1.

Symbolically, $M = (\{q_1, q_2\}, \{1\}, \{1, b\}, \delta, q_1, b, \{q_1\})$

where δ is defined by the following transition table

P.S	Tape Symbol 1	b
~~q_1~~	~~q_2~~	~~q_2~~
$\rightarrow q_1$	bRq_2	–
q_2	bRq_1	–

~~Example~~: Obtain the computation sequence of 11.

$q_1 11b \vdash bq_2 1b$

$\vdash bbq_1 b$.

As there is no movement from $q_1 b$ so machine halts. but string is accepted as q_1 is the final state.

Obtain the computation sequence for 111.

$q_1 111b \vdash bq_2 11b$

$\vdash bbq_1 1b$

$\vdash bbbq_2 b$.

As there is no movement from $q_2 b$ so machine halts but string is not accepted by M as q_2 is not the final state.

Scanned with CamScanner

Example : Design a TM to recognise all string containing of odd no. of 1's.

Sol^n: The construction of is made by defining moves in the following manner :

a) q_1 is the initial state, M enters the state q_2 on scanning 1 writes b.

b) If M is in state q_2 and scans 1, it enters q_1 and writes b.

c) q_2 is the only accepting state.
So M accept the string as it exhaust all the input symbol and finally is in state q_2.

Symbollically, $M = (\{q_1, q_2\}, \{1\}, \{1, b\}, \delta, q_1, b, \{q_2\})$
where δ is defined by the following transition table:

P.S	Tape symbol 1	b
$\rightarrow q_1$	bRq_2	-
q_2	bRq_1	-

Obtain the computation sequence for 11

$q_1 11b \vdash bq_2 1b \vdash bbq_1 b$

As there is no transition from q_2 so the machine halts and string is not accepted as q_1 is not final state.

Obtain the computation sequence for 111.

Example: Design turing machine for $L = \{a^n \mid n \geq 1\}$.

Sol^n: The construction is made by defining moves in the following manner.

a) q_1 ~~ao~~ is in the initial state, M enters the state q_2 on scanning a writes b.

b) If M is in the state q_2 and scans a, it ~~enters~~ remains at state q_2 and writes b.

c) q_2 is the only accepting state.

So M accept the string as it exhaust all the input symbol and finally is in state q_2.

Symbolically $M = (\{q_1, q_2\}, \{a\}, \{a, b\}, \delta, q_1, b, \{q_2\})$.

where δ is defined by the following table.

PS	Tape symbol a	b
$\rightarrow q_1$	bRq_2	—
q_2	bRq_2	—

Obtain the computation sequence for i) aa ii) aaa.

i) $q_1aab \vdash bq_2ab \vdash bbq_2b$

M halts as there is no transition from q_2b. but the string is accepted as q_2 is final state.

ii) $q_1aaab \vdash bq_2aab \vdash bbq_2ab \vdash bbbq_2b$

M halts as there is no transition from q_2b but the string is accepted as ~~there~~ q_2 is final state.